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Cohesive Zone Modeling of the Polyimide/Silicon Interface in 2.5D Packaging by Nanoindentation Testing

The polyimide (PI)/silicon (Si) is a critical interface in 2.5D packaging and is highly susceptible to failure due to thermal and mechanical loads. The large thermal expansion mismatch that exists between these two materials poses a challenge for interfacial modeling and reliability predictions. This study presents a method for extracting and calibrating the cohesive zone model (CZM) parameters of the PI/Si interface by nano-indentation testing. Specimens were designed and fabricated to induce interfacial shear deformation during indentation, and the displacement-load (D-L) curves were measured to derive the preliminary CZM parameters. Under the assumption of a bilinear traction-separation law, the key parameters—initial stiffness K_0 , maximum shear strength τ_{max} , and critical separation displacement δ_f —were extracted. The finite element simulations were then performed using ABAQUS to reproduce the experimental D-L response, δ_f and τ_{max} were further calibrated to improve the agreement between simulation and experiment. The final identified CZM parameters are: $K_0 = 3.6 \times 10^{14}$ Pa/m, $\tau_{max} = 113$ MPa, and $\delta_f = 4.2$ μ m. The simulation results demonstrate that the proposed method enables an accurate characterization of PI/Si interfacial behavior and is able to provide a reliable parameter set for future interfacial fracture simulations. [DOI: 10.1115/1.4070886]

Keywords: 2.5D packaging, PI/Si interface, cohesive zone model, nanoindentation, finite element simulation

1 Introduction

The continued miniaturization and integration of electronic packaging have led to increasing complexity in material systems and interfacial structures. Interfaces between dissimilar materials—such as polymers, metals, and dielectrics—are particularly prone to delamination and crack propagation under thermomechanical loads during fabrication, assembly, and operation, posing one of the

critical challenges to the reliability of micro-electronic and power electronic packaging [1–5]. Coupled multiphysical effects under actual service conditions—such as thermal cycling and power loading—further accelerate interfacial degradation [6–8].

Conventional linear elastic fracture mechanics approaches face inherent limitations in capturing crack initiation and discontinuous propagation in complex interfaces [9–11]. In recent years, the cohesive zone model (CZM) has been widely adopted for interfacial modeling in electronic packaging due to its capability of naturally describing interfacial slip and damage evolution via traction-separation relationships [12,13]. For example, Fei et al. [14] investigated crack propagation at the Cu/SiO₂ interface in TSVs using CZM; Chen et al. [15] extracted CZM parameters for the Cu/

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polyimide (PI) interface in RDL structures; Wu et al. [16] determined CZM parameters for TSV/Si interfaces via nano-indentation-based experiments. These studies demonstrate the advantages of CZM in modeling interfacial fractures in heterogeneous systems.

Reliable acquisition of CZM parameters is critical to ensuring predictive accuracy. Prior studies have addressed this by coupling experiments with inverse methods: Reuther et al. [17] extracted CZM parameters for metal/ceramic interfaces via nano-indentation and inverse modeling; Wu et al. [18] established a CZM model for the adhesive layer of ultrathin chips through micropeeling tests; Mei-Ling Wu et al. [19] proposed a failure prediction framework for interfacial microcracks in MEMS packaging. However, for critical intermediate interfaces such as PI/Si, direct experimental extraction and calibration of CZM parameters have been limited, and many simulations still rely on estimation or simplified hypotheses or even ignore this interface [20–24].

The PI/Si interface is an important structural component in the interposer of 2.5D packaging. Notably, the coefficient of thermal expansion (CTE) mismatch between PI (30.5 ppm/°C) and Si (2.8 ppm/°C) induces significant thermal deformation incompatibility [15], making the PI/Si interface particularly susceptible to warpage and interfacial sliding failures under thermal loads. However, it has received limited experimental attention, and existing studies rarely provide quantitative traction–separation parameters based on mechanical tests specifically designed for this interface.

To fill this gap, this study conducts nano-indentation experiments on customized specimens that introduce interfacial shear loading. Based on the displacement–load responses, key parameters of a bilinear cohesive zone model—including initial stiffness, maximum shear strength, and critical separation—are extracted. These parameters are further calibrated using finite element simulations, which aim to reproduce the experimental curves. More importantly, the proposed method addresses a key limitation of conventional nano-indentation-based CZM characterization, where the indenter displacement is often assumed to be equal to the interfacial sliding. In contrast, our approach explicitly accounts for the elastic deformation of the PI layer and reconstructs the nano-indentation process through finite element simulation, enabling a more realistic representation of the interfacial shear behavior.

The final parameter set offers a reliable foundation for modeling the mechanical behavior of the PI/Si interface. This work presents a practical, test-informed method for interfacial parameter identification and provides essential data for future studies on interface reliability in heterogeneous electronic packaging systems. By integrating experimental displacement–load (D–L) data with simulation-based inverse parameter extraction, the method yields high-fidelity CZM parameters that can be directly applied to thermo-mechanical simulations of 2.5D packaging and offers enhanced predictive capability for multilayer heterogeneous interface reliability.

2 Nano-Indentation Experiment

2.1 Specimen Preparation. The test specimens in this study were designed based on the interlayer configuration commonly used in 2.5D packaging, with the primary purpose of characterizing the mechanical bonding performance of the PI/Si interface. To ensure controllable interfacial stress states during testing and to improve the repeatability of the measurements, interface samples of various sizes were fabricated for nano-indentation and interfacial shear loading experiments.

Femtosecond laser cutting technology was employed for sample preparation. This method offers excellent machining precision and minimizes the heat-affected zone. During processing, the single-pulse laser energy was controlled at approximately 5 μ J, and the scanning speed was set to 20 mm/s to achieve a balance between interface integrity and minimal thermal damage. The final cutting accuracy was maintained within 1 μ m, effectively avoiding microcracks and stress concentrations that are commonly introduced by conventional mechanical or long-pulse laser cutting methods.

Three interface sizes were designed for the experiments: $0.1 \times 0.1 \text{ mm}^2$, $0.1 \times 0.15 \text{ mm}^2$, and $0.1 \times 0.2 \text{ mm}^2$, representing different loading areas for assessing the mechanical response of the PI/Si interface. For each size, three specimens were prepared, resulting in a total of nine sets of test samples. After cutting, all specimens were subjected to ultrasonic cleaning to remove residual particles.

As shown in Fig. 1, the machined interfaces are flat, with clearly defined edges of the retained PI blocks and no obvious burrs, indicating that the femtosecond laser processing had minimal impact on the structural integrity of the interface. This ensures reliable conditions for subsequent high-precision interfacial mechanical testing.

2.2 Nano-Indentation Test. Nano-indentation experiments were conducted using a Bruker Hysitron TI 950 (Eden Prairie, MN) to investigate the interfacial mechanical behavior of the PI/Si interface. The specimen was mounted on a glass slide using a cyanoacrylate adhesive to ensure stable fixation and to prevent the specimen from tipping over during the testing process. The choice of glass as the mounting substrate was intentional: glass possesses a relatively high elastic modulus (of the order of several tens of GPa), which effectively minimizes substrate-induced deformation during indentation and thereby reduces the influence of substrate compliance on the measured response. This consideration is consistent with prior studies emphasizing the sensitivity of nano-indentation results to substrate stiffness, such as the work by Gomez and Basaran [25]. As shown in Fig. 1(b), the specimen was positioned such that the indenter applied load perpendicularly to the direction shown in the figure, thereby inducing shear loading parallel to the PI/Si interface.

During testing, D–L curves were recorded for each sample to enable the subsequent extraction of cohesive zone model (CZM) parameters. All experiments were performed at room temperature in a vibration-free environment to ensure stability and accuracy. A constant displacement rate of 10 nm/s was applied, which was determined through preliminary tests using samples with an interface size of $0.1 \times 0.1 \text{ mm}^2$ to ensure suitable loading conditions for the onset of interface sliding or peeling. To prevent potential damage to the equipment caused by specimen collapse after a sudden load drop, the load was unloaded at an appropriate time after the sudden drop in load.

Figure 2 shows the scanning electron microscopy (SEM) image of the specimen after testing, where the PI block was completely detached from the Si substrate. The clear separation at the PI/Si interface without noticeable residuals or damage indicates that the measured D–L curves reliably reflect the delamination behavior of the PI/Si interface.

2.3 Experimental Data. Under the displacement-controlled loading protocol at a rate of 10 nm/s, nano-indentation tests were performed on the top surface of the PI block. Figure 3 shows the D–L responses of three specimens with an interface size of $0.1 \times 0.15 \text{ mm}^2$. It is noted that valid data could not be obtained from the three specimens with an interface size of $0.1 \times 0.2 \text{ mm}^2$. When the applied load was below 0.8 N, the load and displacement exhibited a linear elastic relationship, indicating that the PI block remained elastic without crack initiation or growth. As the load exceeded 1 N, a slight nonlinear response appeared, which could be attributed to plastic deformation of the compressed PI block, crack initiation, or a combination of both. Notably, under displacement-controlled loading, the load began to decrease after reaching a peak value of approximately 1.39 N, which is likely associated with stable crack propagation along the interface. The final stage of the curve represents the D–L response during unloading.

3 Extraction and Validation of Cohesive Zone Model Parameters

3.1 Cohesive Zone Model. The CZM model is widely used to simulate interface damage mechanisms, such as crack initiation and

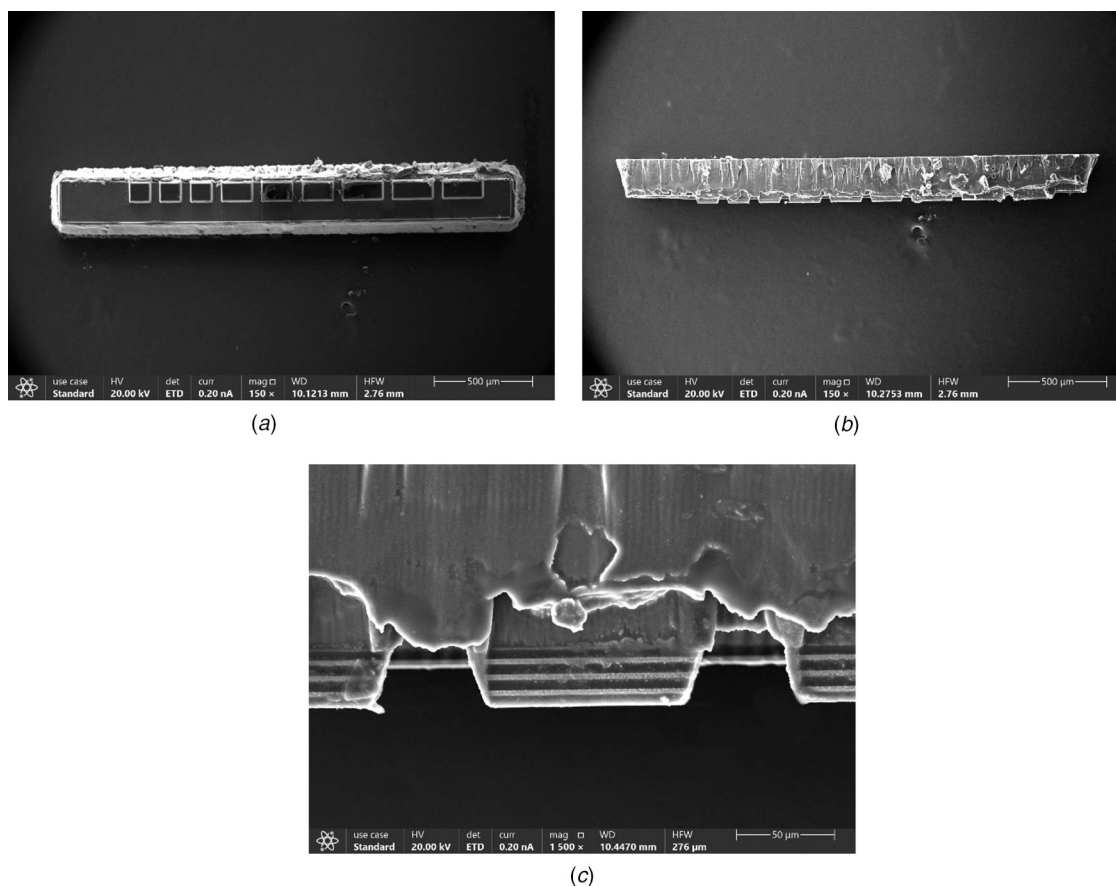


Fig. 1 Nano-indentation test specimen: (a) surface view of the specimen after femtosecond laser cutting, (b) cross-sectional view of the specimen, and (c) enlarged detail of the interface region in the cross-sectional view

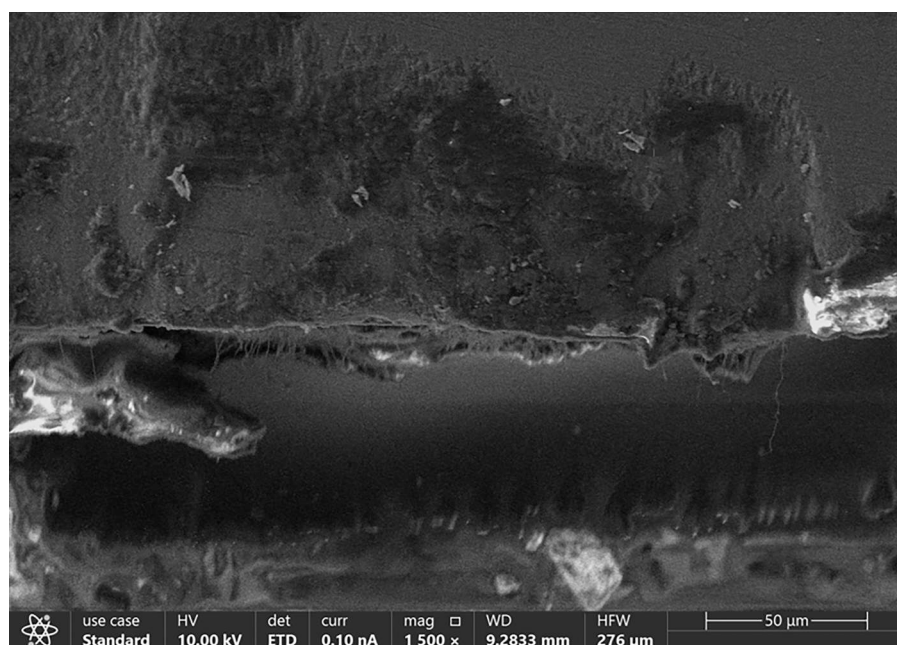


Fig. 2 SEM micrograph of the PI/Si interface after testing

propagation, especially when the material interface is subjected to mechanical and thermal loads [26–29]. These models describe the mechanical response of the interface through the nonlinear relationship between traction and separation forces, which is

referred to as the traction-separation law. Various forms of this law have been proposed in the literature, including perfect plasticity [26], smooth nonlinear [27], trapezoidal [28], and bilinear or triangular models [29]. It is worth noting that the classical CZM

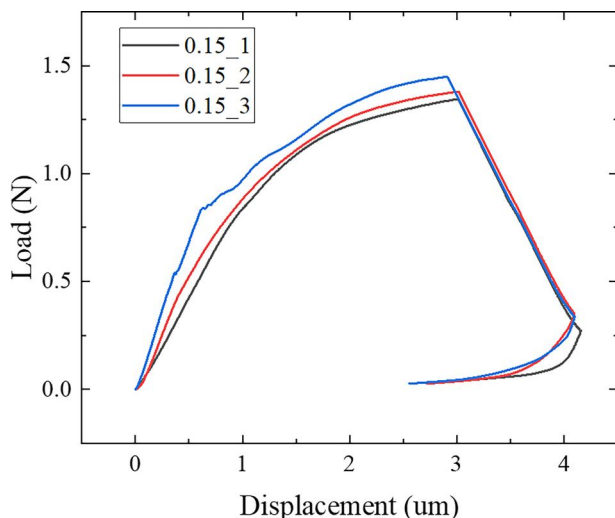


Fig. 3 Representative D-L responses of PI/Si specimens

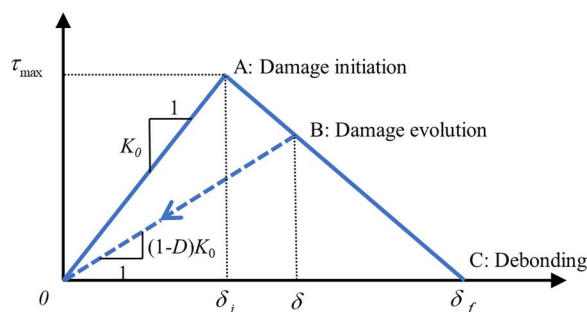


Fig. 4 Diagram of bilinear traction-separation

formulations adopted in most micro-electronic packaging studies, including the present work, are primarily mechanical in nature and do not explicitly incorporate interface heat generation or entropy-based damage mechanisms.

This study employs the bilinear traction-separation law (Fig. 4) to describe the mode II (shear-dominated) debonding process occurring during PI block compression. This law describes interface shear behavior as a function of tangential displacement [30]. In the initial stage, the interface exhibits an elastic response, with shear stress increasing linearly with displacement, controlled by the initial stiffness K_0 . Stress reaches a peak τ_{max} at the critical displacement $\delta_i = \tau_{max}/K_0$. Thereafter, damage begins and develops, causing shear stress to gradually decrease until complete separation occurs at displacement δ_f . Subsequently, stress drops to zero and remains constant, indicating that the interface has completely debonded [31].

The area enclosed by the traction-separation curve quantifies the energy required for complete fracture of the interface, commonly referred to as fractured energy or interface toughness

$$\Gamma = \frac{1}{2} \tau_{max} \delta_f \quad (1)$$

To characterize the progressive degradation of interfacial strength over time, a damage variable D is introduced when the separation displacement δ exceeds the initiation value δ_i but remains below the final value δ_f . This variable ranges from 0 (undamaged) to 1 (fully damaged) and is defined as a function of the maximum separation displacement δ_{max} experienced during loading

$$D = \frac{\delta_f(\delta - \delta_i)}{\delta(\delta_f - \delta_i)} \quad (2)$$

It effectively scales the stiffness of the interface: the current shear traction is given by a modified linear relation with reduced stiffness

$$\tau = (1 - D)K_0\delta \quad (3)$$

Combining Eqs. (2) and (3) give that the shear traction decreases linearly with the displacement δ for $\delta_i < \delta < \delta_f$

$$\tau = \tau_{max} \frac{\delta_f - \delta}{\delta_f - \delta_i} \quad (4)$$

Mathematically, during the damage phase, the shear stress decreases linearly from τ_{max} to zero as the separation displacement δ increases from δ_i to δ_f . During unloading, δ_{max} remains unchanged, and thus damage variable D also remains constant, reflecting the irreversibility of interfacial damage. As a result, the unloading path follows a straight line with a slope of $(1-D)K_0$, as indicated by the dashed line in Fig. 4.

While this classical mechanical description effectively captures crack initiation and separation processes, it does not account for additional thermodynamic contributions—such as heat generation or irreversible entropy production—that may arise at highly stressed or sliding interfaces. In addition, classical CZM formulations typically trigger damage only when the relative displacement reaches a critical initiation value. This displacement-based criterion does not represent subcritical cyclic sliding or the continuous entropy generation required by the second law of thermodynamics. As highlighted in recent advances such as the unified mechanics theory [32], any deformation—regardless of its magnitude—produces irreversible entropy and thus contributes to progressive material degradation. The present work adopts the conventional displacement-based CZM formulation due to its widespread use and practicality in micro-electronic interface modeling, while acknowledging that thermodynamic-based damage descriptions fall outside the scope of this study but represents an important direction for future extension.

3.2 Cohesive Zone Model Parameters of Polyimide/Silicon Interface. In the analysis of the nano-indentation test data for the PI block, the indenter is assumed to be a rigid flat punch that applies a uniform normal stress and end displacement (Δ) on the top surface of the PI block. The PI block is assumed to behave as a linearly elastic material on the surface of a rigid silicon substrate. For each infinitesimal element of the PI block (Fig. 9(b)), the difference in axial stress (σ_x) between the two cross sections is balanced by the interfacial shear stress (τ) arising from the interaction between the PI block and the silicon substrate. The force equilibrium of the infinitesimal element requires that

$$\tau = -\frac{B d \sigma_x}{dx} \quad (5)$$

where B is the thickness of the PI block, and the axial stress in the PI block is related to its axial displacement (u_x) through the linear elastic Hooke's law

$$\sigma_x = E_x \frac{du_x}{dx} \quad (6)$$

Where the effective elastic modulus in the x -direction (E_x) of the multilayer PI/Cu structure was calculated based on the rule of mixtures (Voigt model) as [33]

$$E_x = \frac{\sum_{i=1}^n E_i \cdot t_i}{\sum_{i=1}^n t_i} \quad (7)$$

where E_i and t_i are the Young's modulus and thickness of the i th layer (either PI or copper) in the multilayer structure of the PI block, as illustrated in Fig. 5(b).

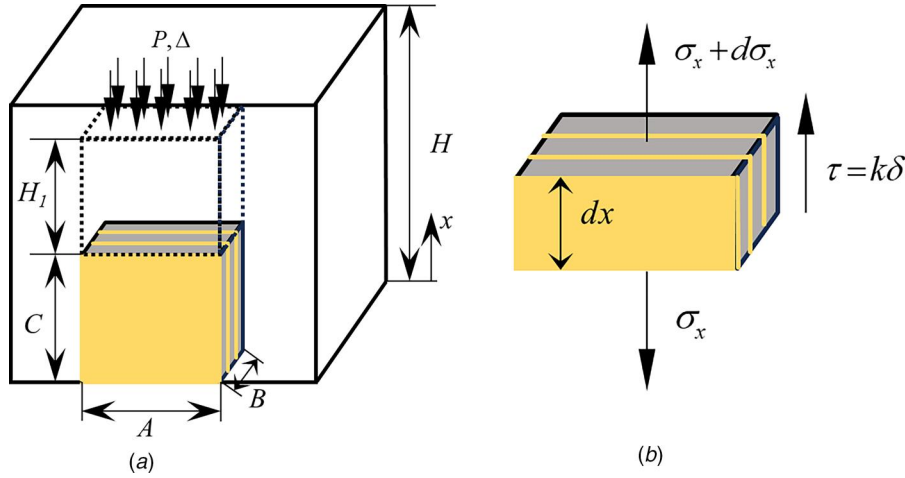


Fig. 5 Schematic of nano-indentation and stress analysis model: (a) nano-indentation test setup schematic and (b) PI block element of length dx with stress and shear force definitions

The relative displacement at the interface is given by $\delta = u_m - u_x$, where $u_m \approx 0$, based on the assumption that the silicon substrate behaves as a rigid material. A shear stress–separation relationship exists between the interfacial shear stress and the relative displacement. Inspired by the measured load–displacement response, the PI/Si interface is assumed to follow a bilinear shear stress–separation law (Fig. 4), which consists of an initial linear elastic stage followed by stable crack propagation, as expressed by

$$\tau = \begin{cases} K_0 \delta, & \delta \leq \delta_i \\ \tau_{\max} \frac{\delta_f - \delta}{\delta_f - \delta_i}, & \delta_i \leq \delta \end{cases} \quad (8)$$

When the end displacement Δ is less than δ_i , the relative displacement δ at any position remains smaller than δ_i , and thus the traction–separation relationship is in the linear elastic stage ($\tau = K_0 \delta$). Under this condition, substituting Eqs. (6) and (8) into Eq. (5) yields (where $\delta = -u_x$)

$$\delta = \frac{BE_x}{K_0} \frac{d^2 \delta}{dx^2} \quad (9)$$

By applying the appropriate boundary conditions, Eq. (8) can be solved to yield

$$\delta = \Delta \frac{\cosh\left(\frac{x}{\lambda}\right)}{\cosh\left(\frac{H}{\lambda}\right)} \quad (10)$$

where $\lambda = \sqrt{BE_x/K_0}$ the characteristic length, and H is the initial height of the specimen. Two boundary conditions are applied: the prescribed end displacement at the top ($\delta = \Delta$ at $x = H$) and zero axial force at the bottom ($\sigma_x = 0$ at $x = 0$). By substituting Eq. (10) into Eq. (6), the corresponding axial stress in the PI block can be obtained, and the resulting indentation force is given by

$$P = \frac{E_x AB}{\lambda} \tanh\left(\frac{x}{\lambda}\right) \quad (11)$$

Furthermore, based on the experimental data, the maximum displacement δ_f is approximately $4.4 \mu\text{m}$. Combining the experimental data with Eq. (11), the value of K_0 can be determined. Using the maximum experimental load and the interface size, the maximum shear stress $\tau_{\max}$ can also be obtained. Therefore, the key parameters of the bilinear shear CZM for the PI/Si interface, derived from the nano-indentation tests, are summarized in Table 1.

Table 1 Extracted parameters for the traction–separation relation

Specimen	K_0 (Pa/m)	$\tau_{\max}$ (MPa)	δ_i (μm)	δ_f (μm)	Γ (J/m ²)
0.15_1	3.58×10^{14}	89.69	3.01	4.36	195.52
0.15_2	3.60×10^{14}	91.98	3.02	4.41	202.82
0.15_3	3.63×10^{14}	96.58	2.91	4.39	211.99
Mean value	3.60×10^{14}	92.75	2.98	4.39	203.44

3.3 Validation and Calibration of Cohesive Zone Model Parameters. As shown in Fig. 6, a finite element model consistent with the geometric dimensions of the experimental specimen was established in ABAQUS. The bottom of the model (Si substrate) was fixed, and the PI block was modeled as a single solid structure with its elastic modulus set to the effective modulus of the multilayer material, as calculated by Eq. (7). The top surface of the PI block was coupled to a reference point, to which a vertical downward displacement load of $5 \mu\text{m}$ was applied. The interface between the PI block and the Si substrate was modeled using the mean values of the bilinear CZM parameters listed in Table 1.

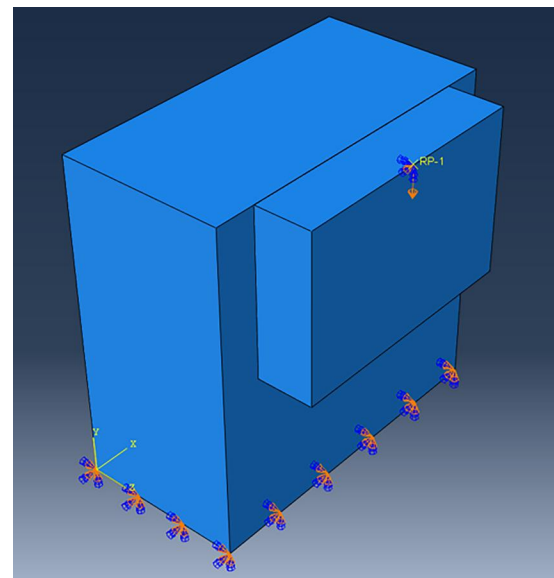


Fig. 6 Finite element model of the nano-indentation test established in ABAQUS

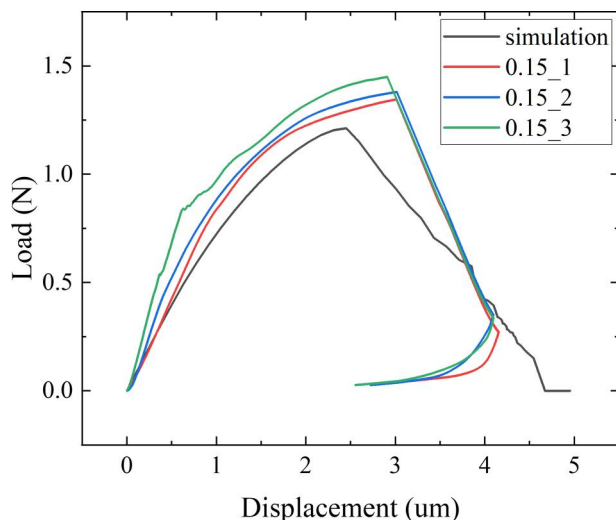


Fig. 7 Comparison between simulated and experimental D-L curves using initially estimated CZM parameters

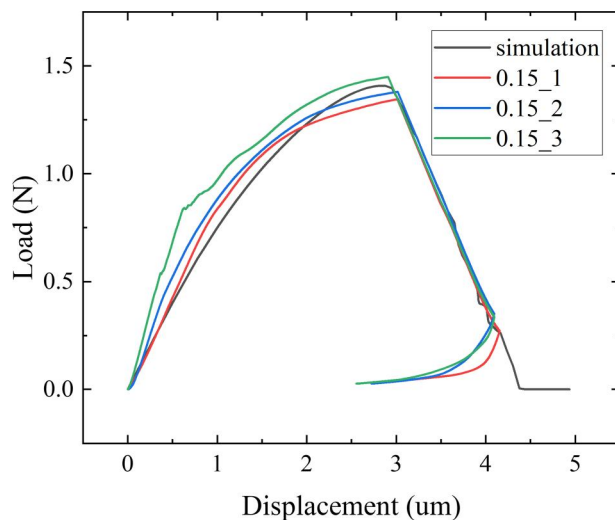


Fig. 9 Comparison between simulated and experimental D-L curves after adjustment of interfacial shear strength $\tau_{\max}$

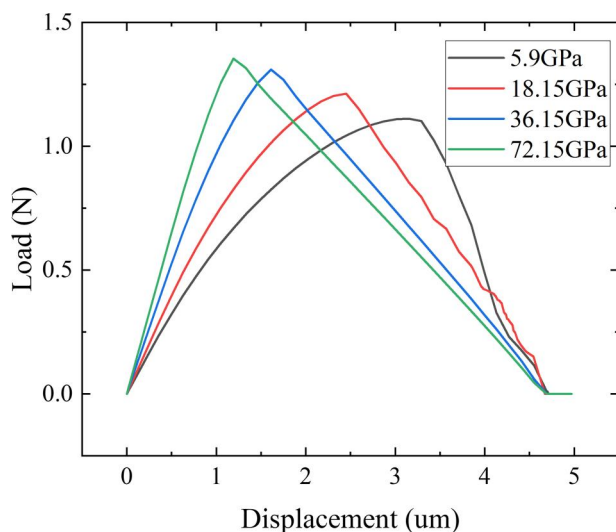


Fig. 8 Effect of PI block elastic modulus on simulated D-L response

As shown in Fig. 7, a discrepancy is observed between the simulated and experimental D-L curves at the peak load on the top surface of the PI block. Specifically, in the estimation of the CZM parameter $\tau_{\max}$, the pressure applied to the top surface of the PI block was approximated as the interfacial shear force. However, prior to interfacial delamination, the PI block undergoes significant elastic deformation, causing the experimentally measured peak load to be lower than the actual critical interfacial load. This introduces a deviation in the estimated $\tau_{\max}$ value.

As shown in Fig. 8, a comparative analysis of the D-L curves obtained under different elastic moduli of the PI block reveals that the simulated peak load increases and gradually approaches the experimental result as the modulus increases. Meanwhile, when the elastic modulus deviates significantly from its actual physical value, the simulated peak load appears earlier than that in the experiment. This phenomenon indirectly validates the earlier analysis regarding the influence of PI block elastic deformation.

Furthermore, in previous studies on TSV/Si interfaces [16], the elastic deformation of the TSV was often neglected due to the significantly higher Young's modulus of copper compared to that of the PI block in this study. Based on the above analysis, it is necessary

to fully consider the effect of PI block elasticity in the present model. Therefore, during the simulation process, the interface shear strength parameter $\tau_{\max}$ was appropriately increased, and the maximum displacement δ_f was reduced. When $\tau_{\max}$ was set to 113 MPa and δ_f was set to 4.2 μm , the simulated D-L curve exhibited good agreement with the experimental results in both overall trend and peak load, as shown in Fig. 9.

4 Application of Polyimide/Silicon Cohesive Zone Model

As shown in Fig. 10, a typical 2.5D packaging model consists of multiple chiplets mounted on the interposer, with interconnections between dies achieved through RDL, while the interposer is further mounted on the package substrate and connected to it via TSVs.

A body heat flux of 1 W was applied to the chiplets as the heat source, with natural convection set on the outer surfaces for steady-state thermal analysis. The temperature distribution results are presented in Fig. 11.

The temperature field from Fig. 11 was applied to the original model as a predefined thermal load for thermo-mechanical coupled analysis. All material parameters in this simulation were referenced from Refs. [15] and [16]. To capture interfacial behavior, CZMs were implemented at key interfaces via surface-based cohesive interactions. Specifically, the PI/Si interface was assigned the CZM parameters obtained in this study, the TSV/Si interface adopted the CZM parameters from Ref. [16] ($K_0 = 4.18 \times 10^{14} \text{Pa/m}$, $\tau_{\max} = 79 \text{ Mpa}$, $\delta_f = 0.189 \mu\text{m}$), and the RDL/PI interface used the CZM parameters from Ref. [15] (normal: $T_{\max} = 45 \text{ Mpa}$, $\delta_f = 3.87 \mu\text{m}$, shear: $T_{\max} = 14.2 \text{ Mpa}$, $\delta_f = 3.17 \mu\text{m}$).

As shown in Fig. 12, the displacement contour after thermo-mechanical coupling analysis indicates a maximum displacement of 11.57 μm , which significantly exceeds the $\delta_f = 4.2 \mu\text{m}$ obtained in this study, demonstrating apparent interfacial sliding failure. In addition, due to the much lower normal strength compared to shear strength in the CZM parameters of the RDL/PI interface [15], minor normal separation occurred between the RDL layer and the PI layer closest to the Si interposer. This phenomenon results from warpage induced by the mismatch in coefficients of thermal expansion (CTE) among heterogeneous materials in 2.5D packaging.

Both tangential interfacial sliding failure and normal interfacial separation can severely undermine the interconnection reliability of 2.5D packaging. The use of accurate CZM parameters enables more precise simulation of thermal deformation in 2.5D package structures, offering significant value for package design.

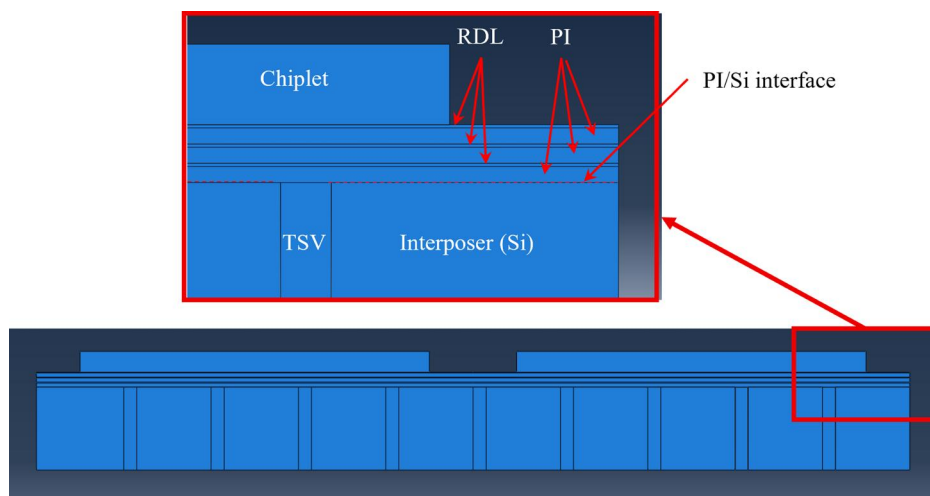


Fig. 10 Geometry of the 2.5D package used for thermomechanical coupling simulation in ABAQUS

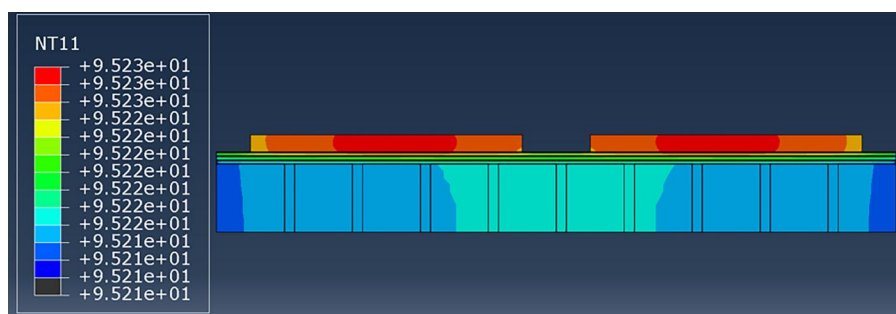


Fig. 11 Steady-state temperature distribution of the 2.5D package under chiplet power loading

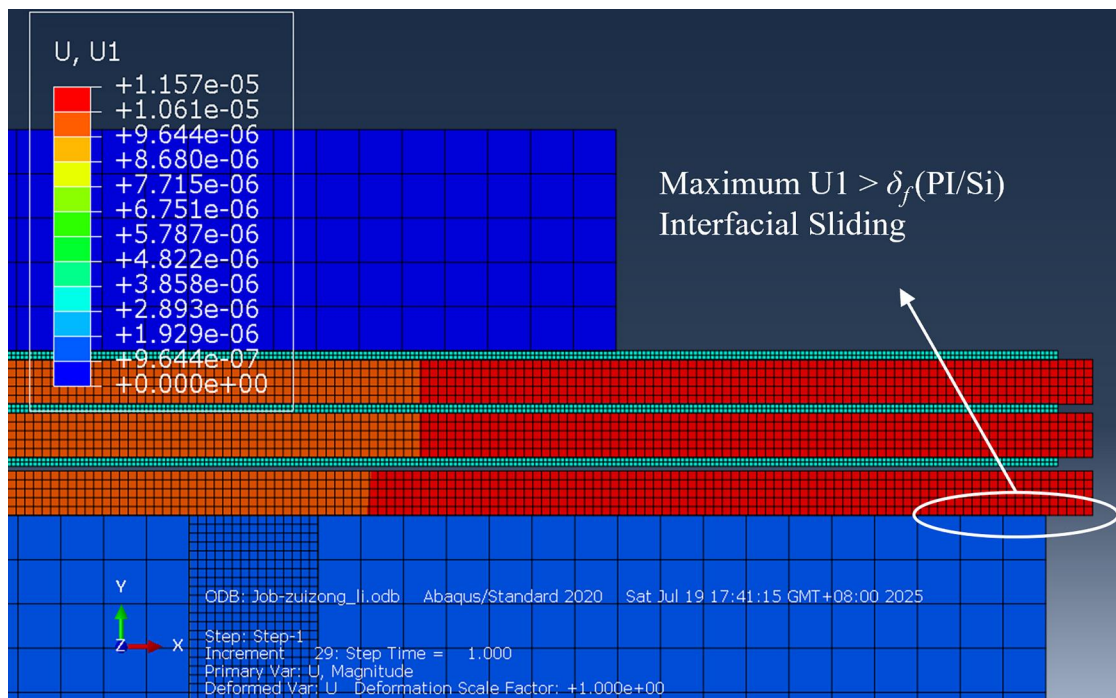


Fig. 12 Displacement field of the 2.5D package after thermomechanical coupling analysis considering interfacial CZM effects

5 Conclusion

In this study, a nano-indentation-based approach, combined with a shear-loading specimen design, is proposed to effectively activate interfacial sliding and delamination behavior, addressing the lack of experimental data for extracting CZM parameters at the PI/Si interface. Finite element simulations are further employed to calibrate the parameters, improving their accuracy and applicability in subsequent modeling. Compared with traditional approaches that approximate the indenter displacement as the interfacial sliding displacement, the present method explicitly accounts for the elastic deformation of the compressed PI layer, enabling the extraction of CZM parameters with higher accuracy and improved suitability for thermo-mechanical simulations of 2.5D packaging.

This study ultimately obtains a set of bilinear CZM parameters for the PI/Si interface that are suitable for numerical modeling: initial stiffness $K_0 = 3.6 \times 10^{14}$ Pa/m, maximum shear strength $\tau_{\max} = 113$ Mpa, and critical separation $\delta_f = 4.2 \mu\text{m}$. Based on the CZM model of the PI/Si interface obtained in this study, together with the CZM models of the TSV/Si and RDL/PI interfaces reported in previous studies, a thermomechanical coupled simulation of the 2.5D packaging structure was conducted. These results provide strong support for interfacial modeling and failure prediction in 2.5D heterogeneous integration packaging structures and lay a solid foundation for further investigation into the mechanical behavior of critical interfaces.

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Nomenclature

CZM = cohesive zone model

D = damage variable

D–L = displacement–load

E_x = the effective elastic modulus in the x -direction of the multilayer PI/Cu structure

K_0 = the initial stiffness of the interface

PI = polyimide

Si = silicon

δ_f = the displacement value when the damage is completed

δ_i = the displacement value at which damage begins

$\tau_{\max}$ = maximum shear stress

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